Campi Flegrei unrest: independent observational results and reproduction record

Independent reproduction of Tan et al. (2025)

12 October 2026

Main paper

Abstract

Earthquake locations, magnitudes and source mechanisms constrain interpretations of unrest at Campi Flegrei. Following the observational study of Tan et al. (2025), this investigation processes original seismic waveforms and historical displacement observations. Independent leveling and GNSS processing recovers part of the displacement history. A limited waveform sample yields nine provisional associated events, without calibrated or double-difference locations. The available results therefore do not establish the publication's fault geometry, magnitude distribution or focal mechanisms. No publication panel is fully independently reproduced. Figure 1B and supplementary S1, S2 and S15C are partial; the remaining panels are explicitly marked not reproduced. The presentation follows the publication's scientific sequence, followed by its supplementary sequence and a separate reproduction record.

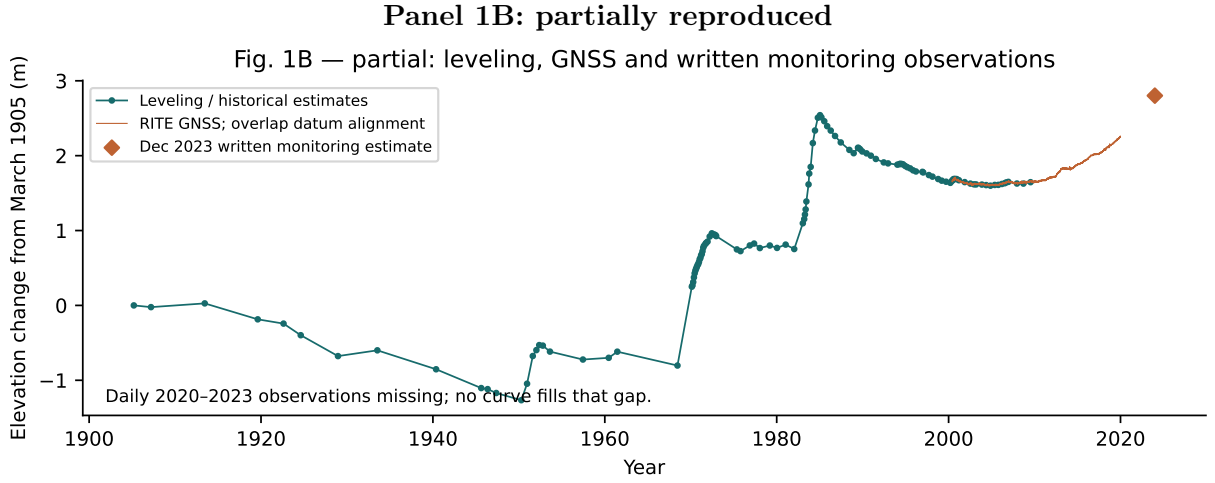
Campi Flegrei's unrest is recorded by ground displacement and seismicity. Tan et al. (2025) use a machine-learning-assisted earthquake catalog to resolve active structures and evaluate possible sources of unrest. Their study is observational: its central results describe the location, timing and character of earthquakes, rather than solutions of a coupled physical forward model. Here, the completed independent calculations are placed in the same scientific sequence. Published images are confined to the separate comparison PDF.

The current phase of unrest

Figure 1B presents the independently processed historical displacement observations. The leveling series contains 128 tabulated entries from 1905–2009, and the daily RITE GNSS series extends through 2019. Relative to the March 1905 leveling observation, the leveling displacement ranges from −1.266 to 2.541 m. The two observation series are joined by a constant offset estimated from their overlap. A separate December 2023 monitoring estimate is displayed as an endpoint; no continuous record is inferred across the missing recent daily observations.

The long record resolves alternating displacement episodes, but its incomplete recent portion cannot establish the detailed timing of deformation during the 2022–2025 seismic study. Panels 1A and 1C remain unreproduced under the existing panel assessment. Separately generated routine-catalog context is preserved in the documentation and is not presented as independent event detection.

Panel A — not reproduced. No valid independent historical seismicity map.



Panel C — not reproduced. No valid independent historical seismicity-rate reconstruction.

Figure 1: Seismicity and ground displacement. Panel 1B is partial: original leveling through 2009 and daily GNSS through 2019, with a separate written December 2023 monitoring endpoint. The vertical datum is relative to March 1905; daily observations for 2020–2023 are missing.

Improving the earthquake catalog

Independent PhaseNet inference and GaMMA association yield nine provisional events in the processed waveform sample. These are initial associations with an alternative starting velocity profile. They have not completed the velocity calibration and double-difference relocation required for a catalog comparable to Tan et al.'s result. Consequently, they cannot support an independent multiyear seismicity rate or depth–time distribution (Figure 2).

All panels/views — not reproduced. Panels A and B require a multiyear calibrated, relocated catalog and magnitudes. Only nine provisional associations exist.

Figure 2: Temporal evolution of seismicity. Seismicity rates by region and depth versus time are not reproduced.

A clearer view of the current phase of unrest

Figures 3 and 4 address the publication's main structural observations: the distribution of seismicity and the source mechanisms of larger events. The present independent work has no calibrated, relocated catalog or focal-mechanism solutions. It therefore cannot determine the depth range, continuity or temporal activation of the ring fault, resolve the central fault system, or establish its slip orientations. The placeholders retain these scientific questions without inserting provisional diagnostic locations or supplied author solutions.

All panels/views — not reproduced. The map and both cross sections require independently calibrated and double-difference-relocated events. Provisional diagnostic locations are not substitutes.

Figure 3: Spatial distribution of seismicity. Map, northing-depth and easting-depth views are not reproduced.

All panels/views — not reproduced. Panels A and B require independent SKHASH/gCAP solutions and reliable locations. No focal inversion was executed.

Figure 4: Focal mechanisms of larger events. Ring-sector and central/onshore focal-mechanism maps are not reproduced.

Implications for the physical process driving the unrest

The reconstructed hybrid-template example in Figure S15C is an independently processed waveform observation. The limited template search produced four unconfirmed candidates; it does not establish recurring hybrid families or their evolution. Neither this result nor the initial associations can test the publication's inference about the presence or absence of upward seismic migration. The partial displacement history also does not distinguish magmatic and hydrothermal forcing. Such an interpretation would require the unresolved spatial and source constraints above.

New perspectives for improved seismic risk assessment

Fault geometry and a calibrated magnitude distribution are needed to connect this analysis to seismic risk. No independently calibrated magnitude was obtained, and the magnitude-frequency analysis in Figure S8 remains unperformed. The current results therefore provide no revised estimate of event probabilities, damaging-event locations or source dimensions. Their supported contribution is restricted to the partial observational reconstructions shown here and in the supplement.

Reproduced supplement

This supplement follows the original Materials and Methods topic sequence, then Figures S1–S15 and Tables S1–S2. It reports only outcomes supported by the saved independent calculations. Detailed processing settings, input inventories, assumptions and validation are collected after this supplement.

Materials and Methods

S1 Building the Campi Flegrei high-resolution earthquake catalog

Data. The independently retrieved station intervals and station inventory provide partial counterparts to Figures S1 and S2. They do not establish complete network coverage over the study period.

Phase picking. Inference with released trained weights produced 3,637 merged picks (560 P and 3,077 S). The training examples and exact benchmark windows in Figures S3 and S4 were not reproduced. Inference on other windows is supporting evidence only.

Phase association. GaMMA associated 63 picks into nine provisional events. These results are insufficient to replace the paper's later location products.

Initial earthquake location. Three events passed the stated geometry screen, but no scientific VELEST inversion was run. Figure S6 and Table S1 therefore have no independent counterpart. The current external routine catalog is contextual evidence for S5, not a replacement for independent locations.

Earthquake relocation. Neither TomoDD relocation nor regional HypoDD SVD uncertainty estimation has been executed. Figures S7 and S9–S13 retain their unreproduced status.

Local magnitude. Of 140 measured channel amplitudes, 23 passed the 5 dB SNR threshold, but no event pair supplied a qualifying magnitude-difference equation. There are no calibrated magnitudes, completeness estimate or b-value for Figure S8.

Focal mechanism estimation. SKHASH and gCAP inversions were not run. No independent mechanism is available for Figure 4.

S2 Validation of earthquake locations

Ring-shaped structures with confined depth distributions. The provisional associations do not establish a ring structure. Internal travel-time consistency is preserved in the documentation, but does not substitute for the regional uncertainty or common-receiver tests in Figures S7 and S12.

Depth of earthquakes in the central region. No associated CPOZ P arrival exists in the processed sample. The specified CPOZ waveform gathers and manual checks on apparently deep central events remain unperformed. No conclusion about a lower depth limit follows from their absence.

S3 Template Matching to Detect Hybrid Events

The search covered 24.298 hours of available common CAAM/COLB intervals and returned four unconfirmed candidates. It did not produce independently confirmed families for Figure S14.

Depth. No independent hybrid-event location set is available for Table S2.

Frequency index. Frequency indices were calculated for the available template and candidate windows. Figure S15C displays the corresponding third written-method template. The first two templates are different examples from the published S15A/B panels and are not inserted in their place.

Relative magnitude. Cluster-relative magnitudes were not assigned because cluster membership has not been established.

Temporal evolution of the hybrid events. The limited exposure and unconfirmed detections do not support a three-year cluster history.

Further search for HB/LP events. Catalog-wide frequency-index and coda-contamination review remains incomplete.

Supplementary figures

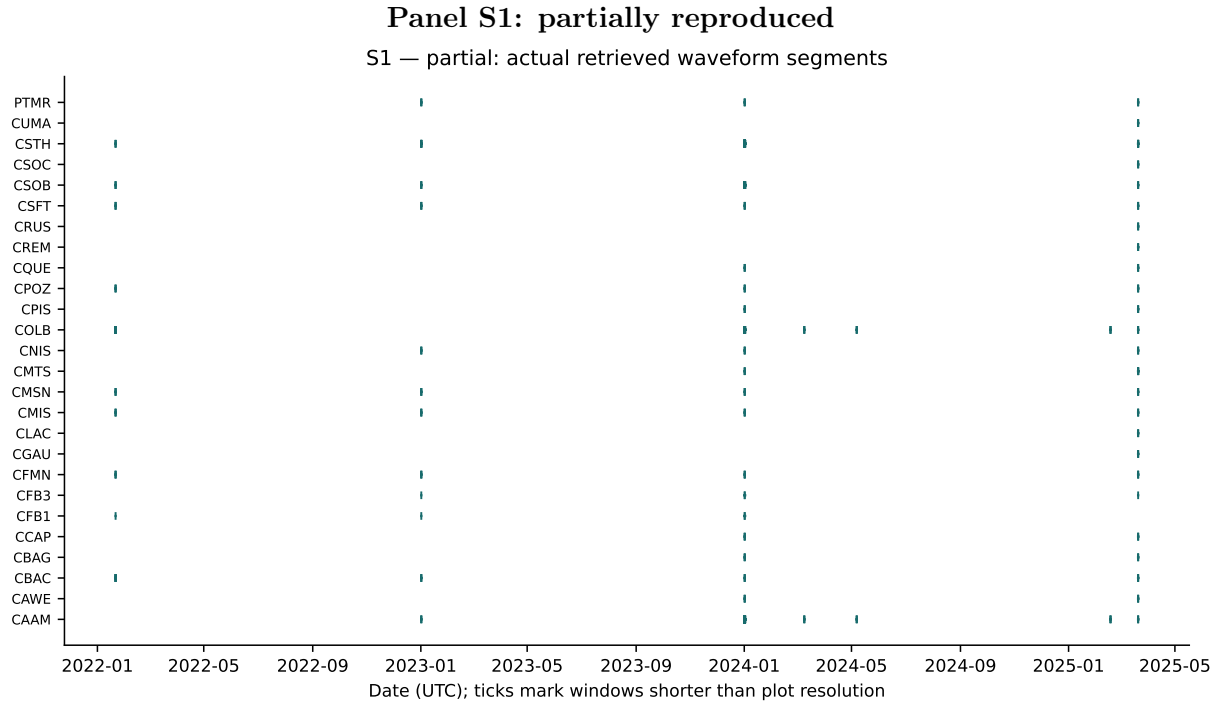


Figure S1: Recorded station availability. Partial: actual returned waveform intervals from 171 files and 26 station names. Short retrieved windows are marked without implying continuous coverage between them; channel-hours are not full network-years.

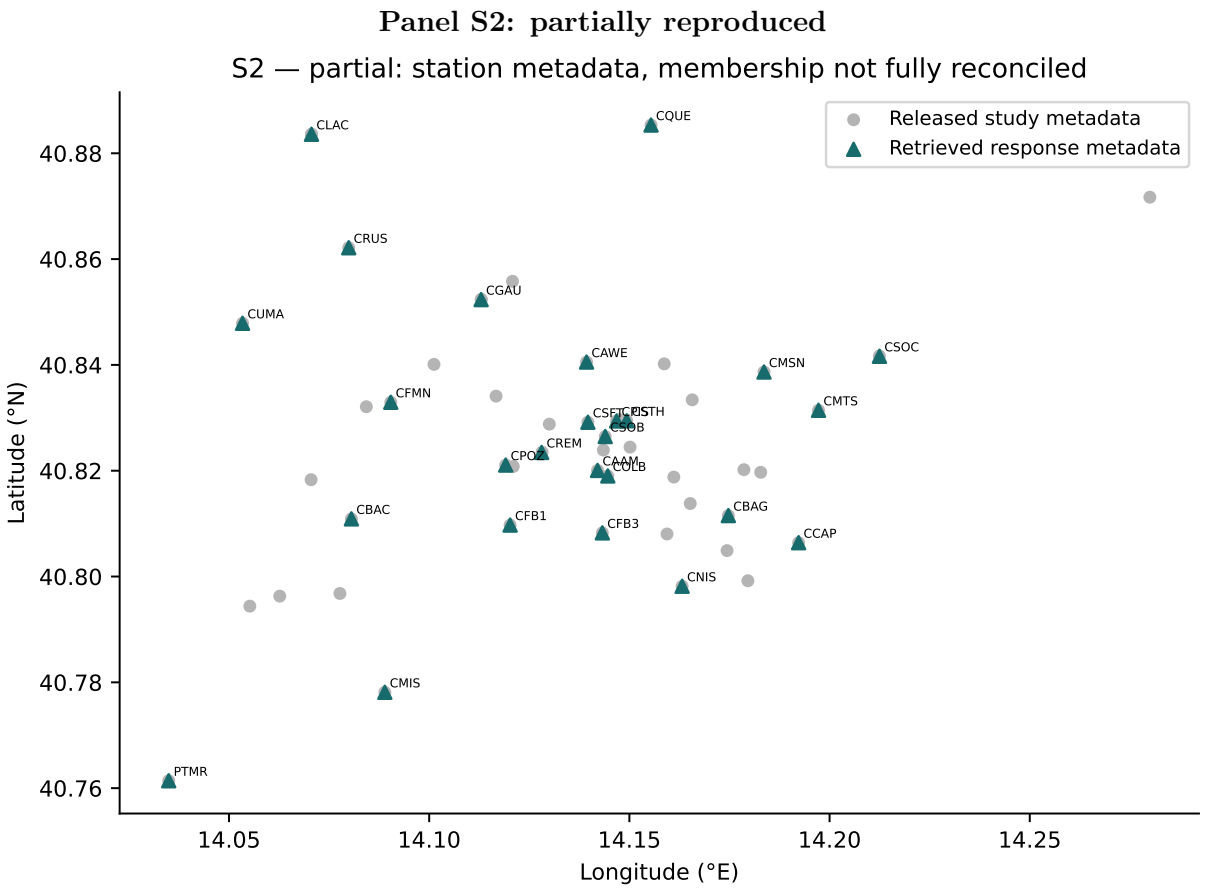


Figure S2: Station locations. Partial: released station coordinates and retrieved IV/2I response metadata. Exact membership of the original 48 instruments, station epochs and the basemap are not fully reconciled.

All panels/views — not reproduced. Panels A–D require the specified training traces, annotations and augmentation examples. These were not reconstructed; downstream use of trained weights is not retraining.

Figure S3: Training examples. Noise, single-event and overlapping-event training examples are not reproduced.

All panels/views — not reproduced. Panels A and B were not generated for the exact CSOB swarm and CMIS noise examples. Other unlabeled inference comparisons remain diagnostics.

Figure S4: PhaseNet benchmark examples. The original benchmark examples and a labeled accuracy comparison are not reproduced.

All panels/views — not reproduced. Current INGV routine-catalog plots are external context. The saved panel record contains no valid independent counterpart to these three views; the published selection/version is unresolved.

Figure S5: Routine-catalog locations. Map and two cross sections retain their not-reproduced status; external catalog plots are documented separately.

All panels/views — not reproduced. The map and both cross sections require scientific VELEST location and velocity/station-correction runs. Compilation alone did not produce these results.

Figure S6: Initial VELEST locations. Initial-location map and cross sections are not reproduced.

All panels/views — not reproduced. Panels A–G require independently selected regional subsets, HypoDD SVD locations and coordinate uncertainties. None of these scientific inversions ran.

Figure S7: Location uncertainties. Regional locations, uncertainty-versus-count panels and uncertainty distributions are not reproduced.

All panels/views — not reproduced. Panels A and B require qualifying relative-amplitude equations and calibrated magnitudes. Zero qualifying equations were obtained; completeness and b-value were not estimated.

Figure S8: Magnitude calibration and frequency distribution. Local-versus-duration magnitude comparison and magnitude-frequency distribution are not reproduced.

All panels/views — not reproduced. All three views require independently relocated events and cross-correlation observation counts for the published quality filter. These outputs are absent.

Figure S9: Best-located earthquake subset. The filtered map and cross sections are not reproduced.

All panels/views — not reproduced. The four annual groups for 2022–2025 require the independent multiyear relocated catalog. The available sample does not support them.

Figure S10: Seismicity by year. All four annual groups are not reproduced.

All panels/views — not reproduced. The four swarm maps require the independent relocated catalog and justified swarm windows. These selections and locations remain incomplete.

Figure S11: Four major swarms. All four swarm maps are not reproduced.

All panels/views — not reproduced. Panels A–H require relocated sector selections and independently associated CPOZ P arrivals. The current associations contain zero CPOZ P arrivals.

Figure S12: Ring sectors and CPOZ common-receiver gathers. Sector views and six arrival-aligned waveform gathers are not reproduced.

All panels/views — not reproduced. Panels A and B require reliable independent relocated coordinates and an appropriately sourced basemap. The relocated catalog does not exist.

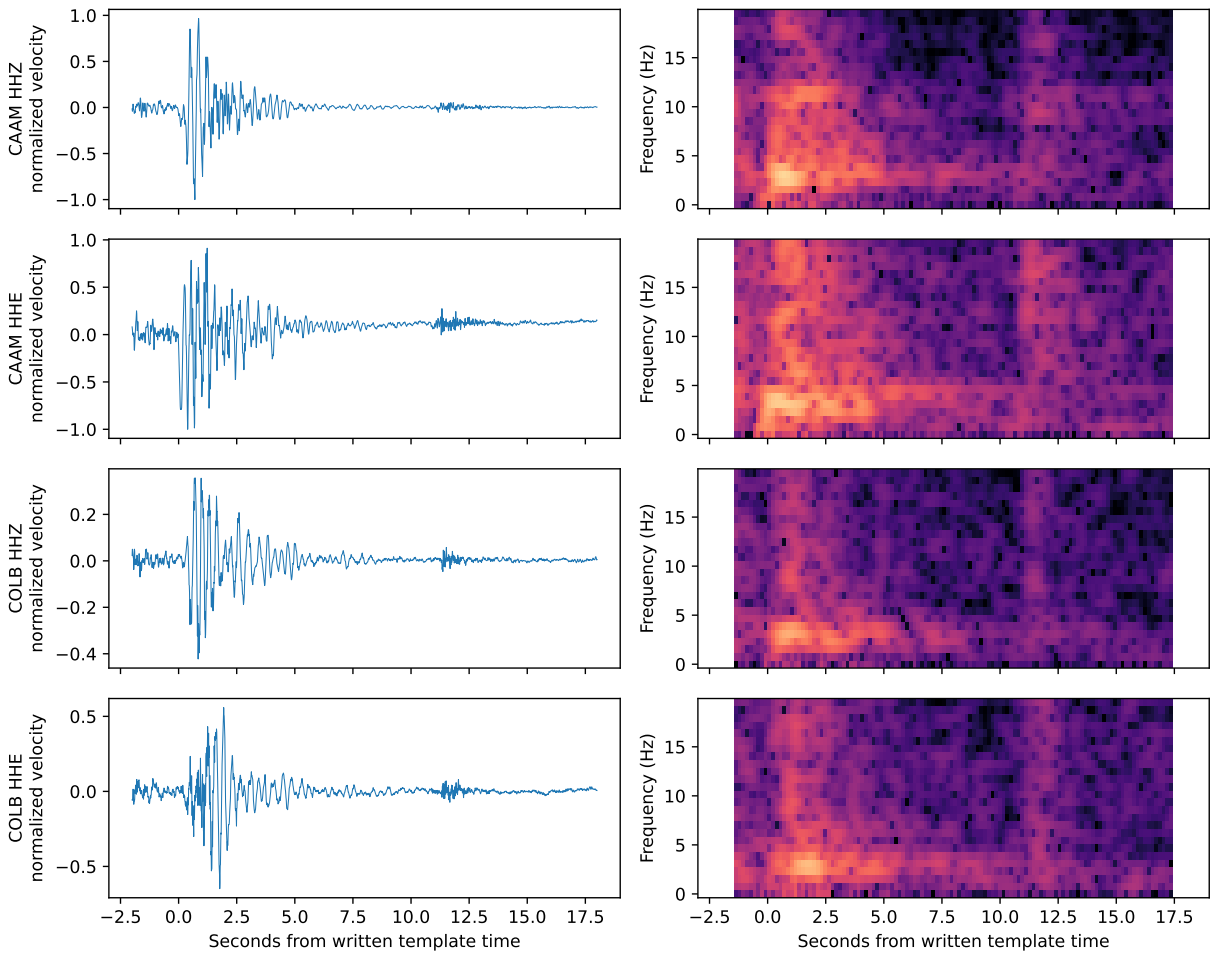
Figure S13: Relocated seismicity on satellite imagery. Full-caldera and onshore spatial views are not reproduced.

All panels/views — not reproduced. The three clusters and unclustered group require a multiyear scan, confirmed events, frequency indices and cluster-relative magnitudes. Four current candidates are unconfirmed.

Figure S14: Hybrid-event clusters. Cluster histories and annotated waveform collections are not reproduced.

Panels A and B — not reproduced. The retained E1/E2 templates are different published examples and are not substituted.

Panel S15C: partially reproduced



Panel D — not reproduced. No independently selected matching shallow-event example.

Figure S15: Hybrid and shallow-event waveforms. Panel C is partial: independently processed E3 CAAM/COLB observations. Onset alignment, display filtering and spectral settings differ or are incompletely specified. A, B and D have placeholders rather than substitute waveforms.

Supplementary tables

Table S1: fitted one-dimensional velocity model — not reproduced. No independently fitted VELEST model exists. The author's tabulated fitted velocities are not copied into this result.

Table S2: hybrid events in the relocated catalog — not reproduced. No independently relocated hybrid-event sample exists. The author's event parameters are not substituted.

Supplementary data products

Independent counterparts of the final catalog (Data S1), associated phases of that final catalog (Data S2), focal mechanisms (Data S3), and the relocated interactive view (Interactive Plot S1) are not complete. The saved provisional products retain their existing labels. References and the complete provenance record follow in the reproduction documentation.

Reproduction documentation

The scientific execution record below describes the saved 9 October 2026 analysis. The 12 October revision reorganizes its presentation and builds a separate visual-comparison document; it does not rerun those scientific calculations. The document-only build verifies that all protected files remain unchanged.

D1 Scientific specification and input verification

The local `tanmodel.pdf` and `tanmodelsupp.pdf` were extracted before changing outputs. Main captions 1–4, supplementary captions S1–S15 and methods S2–S11 are readable. Two font-weight warnings do not prevent main-text extraction. Both supplied ZIPs were listed and CRC-tested; no file from them was executed. Their inventory and hashes are in `docs/resumed_source_check.json` and `docs/file_inventory.json`.

File	Bytes	Inspection
`tanmodel.pdf`	2140554	Text and captions readable
`tanmodelsupp.pdf`	62164947	Text and captions readable
`science.adw9038_data_s1_to_s3.zip`	12960255	Directory readable; CRC passed; not executed
`science.adw9038_interactive_plot_s1.zip`	8852485	Directory readable; CRC passed; not executed

The data archive contains Data S1 (final event catalog), S2 (associated phases) and S3 (focal mechanisms), plus binary macOS packaging metadata. The other archive contains an HTML seismicity visualization. These are derived products, not original waveform inputs. Data S1 was read only after independent outputs existed, for a separate final comparison of temporal coincidences and horizontal differences. Data S3 was read only for a final row-count audit: the CSV has 74 rows and 58 nonempty gCAP solutions, consistent with the methods. Its caption's 55-event description is inconsistent with the actual file. No supplied solution entered inference, initialization, filtering or magnitude estimation.

The latest instruction permits an external input genuinely required by the published method. Accordingly, routine INGV duration magnitudes are permitted only after independent detection and association, as supplementary S6's explicit magnitude regularization reference. Their use no longer presents the blanket policy conflict stated in the previous report. The picking/association program never opens that catalog. Historical catalog plots are explicitly external context, not independently detected earthquakes.

The authors' fitted 1D velocity model, station corrections, picks, associated catalog, focal solutions and study scripts remain excluded. Generic PhaseNet/SeisBench, GaMMA and VELEST are attributed method software. The trained PhaseNet artifact is used for new inference; retraining is not claimed. All independent numerical figures read saved CSV or NPZ outputs. No figure was digitized. Published images appear only as labeled visual references in the separate comparison PDF; none substitutes for an output.

D2 Data access and actual coverage

The frozen rebuild snapshot contains 230 external files (435,600,234 bytes). Original and extended processing read 171 waveform files from 26 stations, representing 428.983 channel-hours after unioning overlapping intervals. This is a lower bound on retrieved coverage, not a full-network availability estimate. The study interval is 21 January 2022 through 20 March 2025. All 26 full-period IV/2I station

requests returned HTTP 413, explicitly citing a one-billion-sample limit. This is a request-size constraint, not evidence that the archive is unavailable. Half-hour requests were then executed for calendar-selected windows on 21 January 2022, 1 January 2023, 1 January 2024 and 20 March 2025. These windows were chosen independently of every earthquake catalog. Daily requests were also attempted; most timed out after 180 seconds, while two returned complete files. Failed transfers are not counted as full-day coverage. A resumable retrieval tool and exact input manifest accompany the project. Full-period download and processing remain unfinished.

Complete response metadata was obtained for 26 station names in networks IV and 2I. This extends the previous four-station response inventory and includes four temporary-network stations. Metadata epochs do not prove continuous recording. The coverage figure uses returned miniSEED segments and preserves gaps. Coverage totals count each channel separately; they do not imply three-component coverage throughout every station interval. Some single-channel records do not support the three-component picker. Actual input files, channel segments, sampling rates, and unioned intervals are saved in the manifests and coverage tables.

The INGV landing page fails local TLS issuer verification, whereas the official FDSN service works with normal verification. RAN's public pages are readable but waveform downloads require registration/login; credentials were unavailable. No account was created. Individual HTTP 204, HTTP 502 and timeout results remain explicit in the receipt files.

The temporary/local ZIP's directory was read using HTTP byte ranges. It lists 319,890 members and approximately 3.845 GB of uncompressed content. Its actual filename station inventory includes CFB1, CFB3, CROS, V0102, V0105, CRUS and CMDP; it does not establish every station advertised on the landing page. One miniSEED member was read successfully, with 6001 samples over 60 seconds. These observations are cut around author-detected events and therefore cannot supply unbiased continuous event discovery. They were not used to seed analysis. The four 2I stations were instead accessed through continuous FDSN requests.

D3 Independent processing and explicit departures

D3.1 Picking and provisional association

The picker uses released tensors with a strict tensor-only load, 100 Hz sampling, nominal 30-second windows, ENZ components, standard-deviation normalization, 1500-sample overlap, 250-sample edge blinding and a 0.3 P/S threshold. The release README specifies a 1 Hz highpass; four poles and a causal implementation are explicit choices. Extended processing uses 30-minute cores with 15-second context where recorded. Missing intervals are not interpolated. Repeated picks within 0.25 seconds at the same station and phase are merged by retaining the larger score. This merging tolerance is an assumption. S picks at CASE, CFB1 and CFB3 are excluded.

The numerical starting profile cited by the paper is Calò and Tramelli (2018). Its retrieved supplement presents that profile graphically; no numerical table was found, and its data statement offers data on request. No plot was digitized. The released fixed 3D file is identified in the model repository as the TomoDD input; the paper attributes its fixed background to Battaglia et al. (2008). Its original-release byte identity remains unverified. The file contains a $23 \times 14 \times 27$ grid of V_P and V_P/V_S; V_S is calculated by division before averaging.

Two independent horizontal averages were computed: 14.02–14.20 degrees E, 40.77–40.86 degrees N, and a central box at 14.11–14.16 degrees E, 40.81–40.84 degrees N. These polygons are explicit assumptions, not inferred figure borders. Bounding nodes and the deep 200 km padding level are omitted from the association profile. Above sea level, the surface velocity is extended to station elevations rather

than averaging air nodes. This profile is a sensitivity alternative; the associated events cannot be called an exact reproduction of the paper's first locations.

GaMMA uses its 1D eikonal travel-time option with 0.1 km grid spacing. The association requires at least six picks, without separate P/S minima. Additional declared settings are a 10-second DBSCAN neighborhood, three minimum DBSCAN samples, fivefold mixture oversampling, a time-only likelihood, 1 second-squared covariance prior, horizontal bounds of ±12 km and depth bounds of 0–8 km. Amplitude is omitted from association because the paper does not specify its association-amplitude configuration; the GaMMA placeholder magnitude is removed from saved events. These choices are in `config/association.json` and were not tuned against a supplied catalog.

D3.2 Location quality and magnitude attempt

Travel-time residuals are recomputed from each provisional origin and the same eikonal model. I calculate

$$R = \sqrt{\frac{1}{N}\sum_{k=1}^{N}(t_k - t_0 - T_k)^2},$$

and the largest unoccupied azimuth between distinct stations. The generic VELEST guidance requires more than six observations and a gap below 180 degrees for calibration candidates. This is a geometry screen, not acceptance of their waveform interpretations. The paper's 0.25-second rejection is a post-location criterion and is not imposed prematurely as a calibration prerequisite.

The generic VELEST source builds after three platform-only repairs: a FORMAT trailing comma, the CPU clock, and the date/time routine. Its scientific inversion was not run. The small provisional candidate set does not constrain the chosen 21-level P/S model, hypocenters and station corrections. The paper used 373 calibration events. No author-fitted model fills this gap. Consequently, the prescribed VELEST residual-rejection sequence, two TomoDD rounds and regional HypoDD SVD uncertainty tests remain unexecuted.

The magnitude attempt nevertheless measures available waveforms independently for all provisional events, without restricting it to the VELEST calibration subset. A recorded P pick defines the arrival, with the calculated P arrival used where only S is assigned. Full response removal produces velocity with a 0.1, 0.2, 35, 45 Hz prefilter and 60 dB water level; these are stated measurement assumptions, not fully specified publication settings. Signal and noise powers use five seconds after and two seconds before the arrival:

$$\mathrm{SNR} = 10\log_{10}(P_{\mathrm{signal}}/P_{\mathrm{noise}}).$$

Observations below 5 dB are rejected. Candidate event pairs require at least two common stations and separation below one-fifth of both source–station distances. The measured equation is

$$M_i - M_j = \log_{10}(A_{is}/A_{js}).$$

The implemented least-squares objective weights channel equations by squared waveform correlation and uses routine M_d priors with declared unit regularization weight. A connected component must contain measured amplitude-difference equations and a matched external anchor. It cannot acquire a magnitude merely because an external M_d exists. This run produced no qualifying edge and therefore no calibrated magnitude. Completeness and b-value cannot be estimated from an empty independently calibrated sample.

D3.3 Hybrid waveforms and search

The three templates use the times written in supplementary S9. Matching uses eight-second CAAM/COLB vertical windows, a causal four-pole 0.2–6 Hz filter, mean two-station normalized

correlation above 0.25, eight-second peak separation and exclusion of ten-second segment edges. Every supported downloaded common interval is scanned, with duplicate exposure removed. This is not a three-year search.

Candidate frequency indices use separate response-corrected, eight-second CAAM windows:

$$\mathrm{FI} = \log_{10}(P_{8\text{–}15\,\mathrm{Hz}}/P_{2\text{–}5\,\mathrm{Hz}}).$$

The matching bandpass is not reused for FI. Candidate waveform panels contain largely persistent broadband fluctuations and do not establish an isolated hybrid onset with a decaying low-frequency coda. All remain unconfirmed. No relative magnitude is assigned to a supposed hybrid cluster before common-source membership is established. The original three template displays remain useful observations; only the E3 example corresponds to S15C. E1/E2 displays are not S15A/B reproductions.

D4 Supporting results, validation and final numerical comparison

Extended inference produced 3,637 merged picks (560 P and 3,077 S). GaMMA provisionally associated 63 picks into 9 events. Only 3 events pass the stated geometry screen; none has completed independent VELEST/TomoDD processing. The magnitude attempt measured 140 channel amplitudes, 23 above 5 dB, and produced 0 qualifying equations and 0 calibrated magnitudes. Post-association external matching found 3 unique temporal matches within two seconds; these did not affect event detection or acceptance. The template search scanned 26 intervals (24.298 hours of valid start times), generating 4 unconfirmed candidates. Analytic association location error was 5.96e-07 km and origin-time error 0 s. A strictly downstream comparison against supplied Data S1 (54,319 events) finds 5 unique temporal matches within two seconds among 9 provisional events. For these matches, the median absolute origin-time difference is 0.350 s and median horizontal difference is 1.042 km (maximum 1.730 km). These differences describe an initial, uncalibrated sample. Temporal proximity alone does not prove event identity, and unmatched events are not declared false detections. The comparison changes no independent data or setting. The association consistency test recovers one analytically constructed homogeneous-medium event with 16 P/S arrivals. It tests association and coordinates, not earthquake-detection accuracy or the adequacy of the alternative background profile. Probability bounds, S-pick exclusions, six-pick minima and assignment uniqueness are checked. The original 27 C++ equation checks and observational waveform/FI checks are retained and rerun. All 27 checks passed. The maximum absolute FI error across the analytic mixtures was 4e-15; the maximum exact-recovery magnitude error was 1.94e-16. These errors compare against analytic fixtures, not observed earthquakes.

An additional inference comparison uses the generic original PhaseNet weights converted by SeisBench and the released retrained weights. It processes calendar-selected CSOB and CMIS observations with the same stated preprocessing. In the 1600-second CMIS window, the original model returns 17 P and 4 S picks, versus 11 P and 55 S picks from the retrained model. The waveform is dominated by persistent fluctuations. Without independently reviewed annotations, these counts establish neither greater sensitivity nor better accuracy. The 120-second CSOB window yields no original-model pick and one retrained S pick. Neither window is claimed to match S4's illustrated example. No retraining or labeled benchmark was executed. Saved probability arrays and both plots remain in the repository. The CMIS example is shown below as a reproduction diagnostic; the CSOB counts are retained here without repeating its figure.

The location-consistency diagnostic compares observed and calculated S–P intervals at stations with both independently assigned phases. It does not validate location accuracy independently of the fitted model. No associated CPOZ P arrival exists in this sample, so the paper's arrival-aligned CPOZ sector gathers cannot yet be formed. Counts and residuals are saved in `data/extended/location_consistency.json`; no author pick fills this gap.

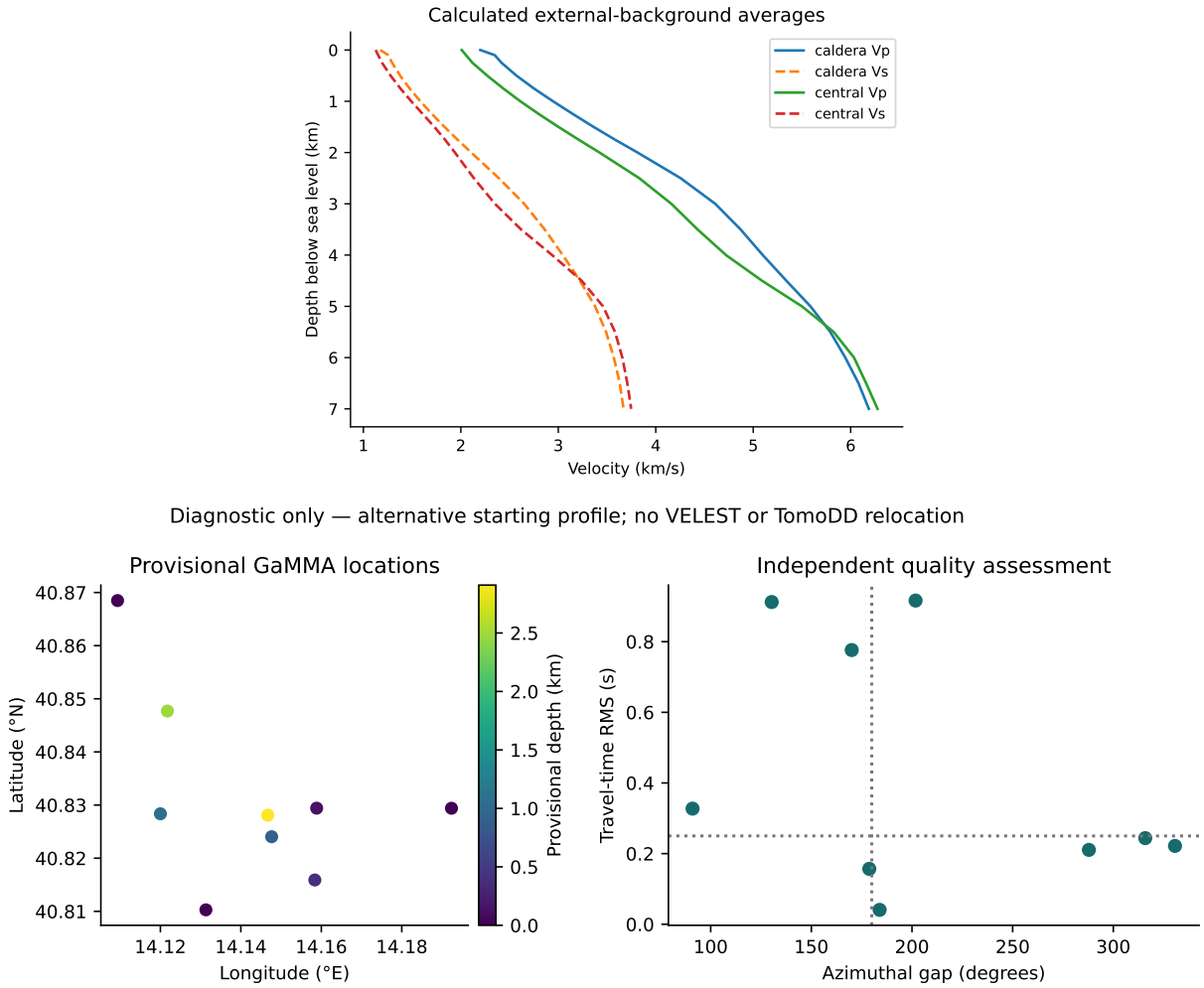
The historical leveling calculation independently extracts 128 tabulated entries from the original Del Gaudio et al. supplement, March 1905–July 2009. It subtracts the March 1905 benchmark elevation. Display dates use the middle of the tabulated month; uncertain months remain flagged in the saved CSV. The resulting displacement ranges from −1.266 to 2.541 m. Some source entries are estimates from photographs or earlier graphical reconstructions; their descriptions are retained. This investigation digitizes no graph. The original De Martino et al. (2021) supplement was subsequently retrieved from the publisher's static content host despite HTTP 403 on its main supplement page. Its RITE daily file supplies North, East and Up positions and uncertainties in meters, with Modified Julian Dates, through December 2019. I estimate a constant datum offset as the median difference between each overlapping leveling observation and the median GNSS Up value within 15 days. Both original series are retained; this assumes a common local displacement history and does not remove spatial differences between benchmarks. The offset and its median absolute deviation are recorded. Original daily observations for 2020–2023 remain missing. The December 2023 official monitoring bulletin gives a written approximate uplift of 1.19 m since November 2005. A separate endpoint is referenced to the median of the original November 2005 GNSS observations; no curve fills the intervening gap and no plot pixels supply values. A later interpolated compilation is excluded as a replacement.

Routine catalog pages for 2005–2025 were retrieved for explicit external context and the permitted calibration role. The current study-period download has 19,274 records, of which 12,503 have location and depth, versus 12,083 events in the publication's S5. This 420-event difference must not be removed by matching an author event list. Possible causes include archive revisions or selection differences; they are not resolved here. Context counts include unlocated events where appropriate, whereas maps require coordinates. The three-year independent detection rate, relocation precision, ring-fault geometry and magnitude-frequency distribution cannot be validated from the limited provisional sample.

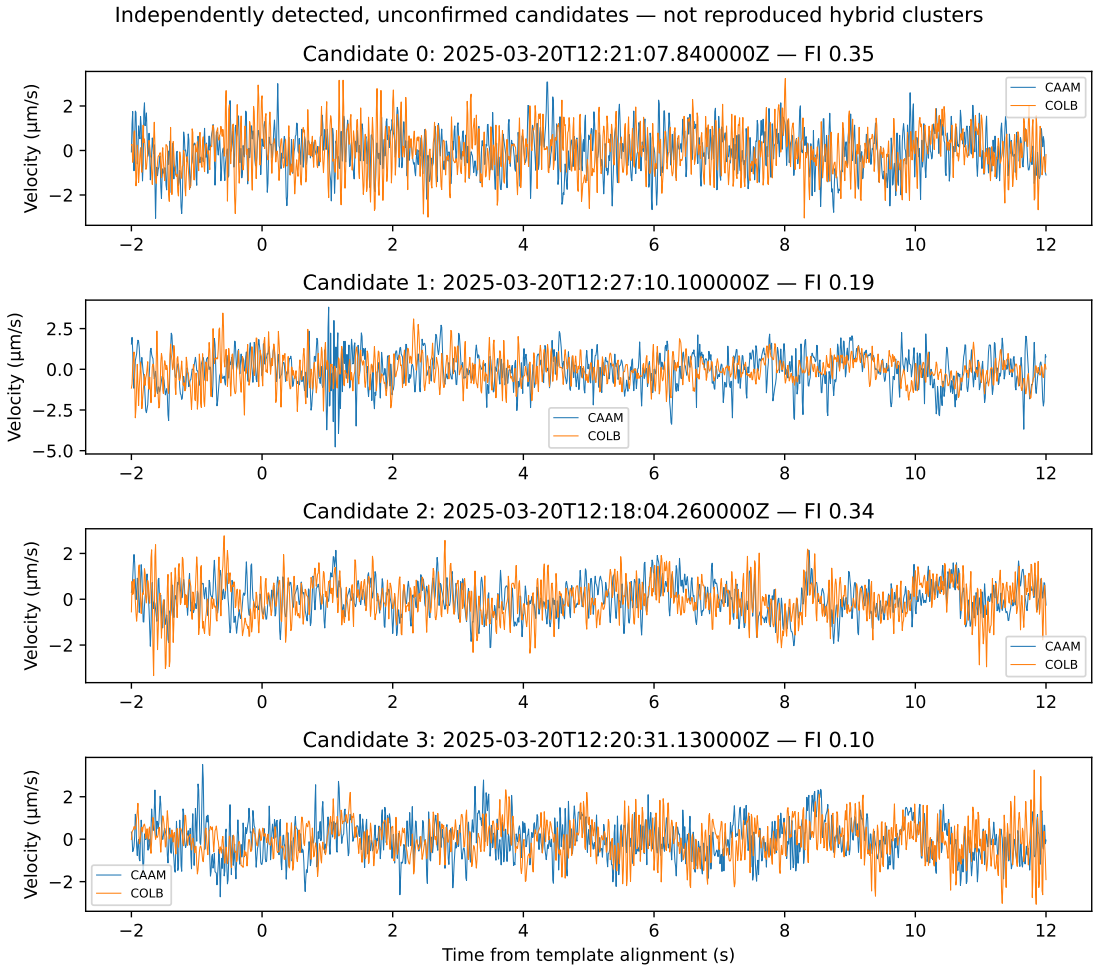
After generating the displacement plot, I inspected the published Figure 1B visually. The sequence of early subsidence, uplift episodes and subsequent relaxation agrees qualitatively. The displayed vertical datum differs: the present plot subtracts the original table's March 1905 benchmark height of 3.464 m, whereas the source elevations are retained in the saved table. Absolute plot ordinates should therefore not be compared without accounting for the datum. No value was digitized or used to adjust the analysis from this comparison. The incomplete recent daily series precludes a full temporal reconstruction.

D5 Essential reproduction diagnostics and external context

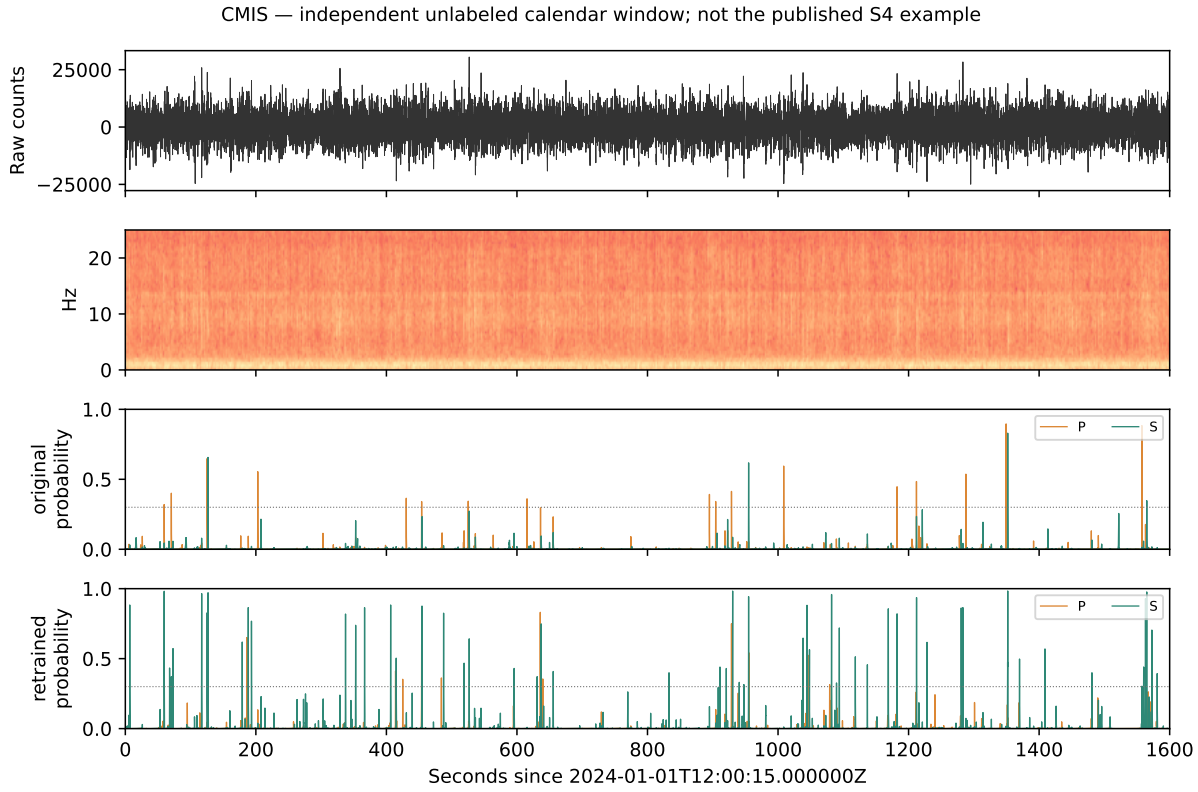
The figures in this section are supporting evidence, not publication figures. They are excluded from both the main-paper/supplement figure sequence and the comparison PDF. Other diagnostic source data and scripts remain in the repository without additional report figures.



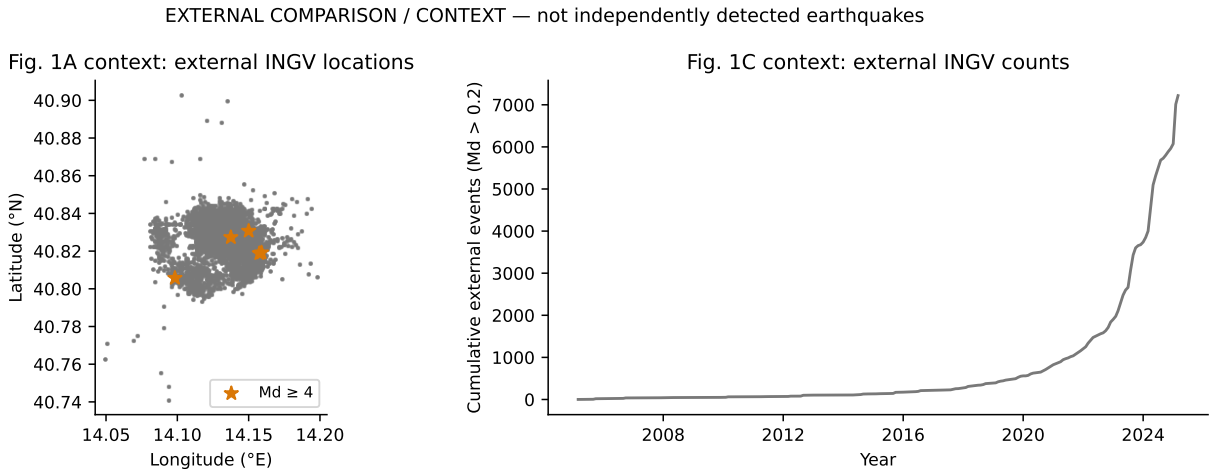
Reproduction diagnostic D1: Alternative background averages and provisional association. The profiles explain the departure from the specified starting model, and the location/residual/gap panels show why an initial diagnostic cannot stand in for Figure 3 or the VELEST/TomoDD products. No calibrated structural result is claimed.



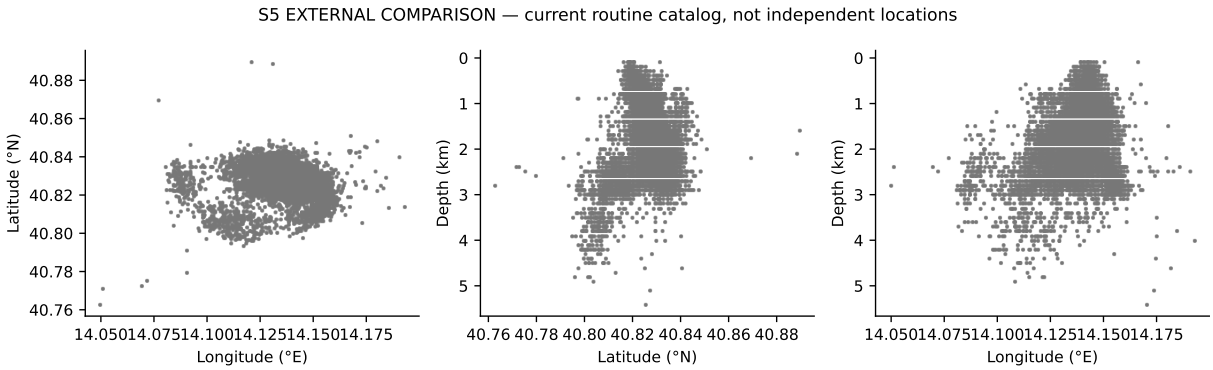
Reproduction diagnostic D2: Unconfirmed template-search candidates. Threshold crossings alone do not establish isolated hybrid onsets or common-source clusters. This evidence supports withholding a reproduced status from Figure S14.



Reproduction diagnostic D3: Original and retrained PhaseNet inference on an unlabeled CMIS calendar window. Different pick counts without reference annotations do not establish comparative accuracy. This is not the published S4B example. The related CSOB arrays, figure and numerical counts remain available in the repository and text.



External context D4: Current routine INGV historical seismicity, related to the questions in Figure 1A/C. These are external catalog records, not independent detections, and retain the existing not-reproduced panel assessment.



External context D5: Current routine-catalog study-period locations, related to S5. The retrieved 12,503 located events differ from the publication's 12,083; neither selection matching nor independent reproduction is asserted.

D6 Completion status and remaining work

No publication panel is fully independently reproduced. Figure 1B and S1, S2, S15C are partial. All remaining main-text and supplementary panels are not reproduced, including those for which an external context plot is supplied. The full record gives each quantity, written specification, exact inputs, executed data/plot commands or null commands, and a specific limitation. Generic software access is distinguished from scientific execution.

Stage	Actual execution and limitation
1. Collection	Executed calendar windows and whole-period requests; 26 stations retrieved. Full-period chunked download remains unfinished. RAN credentials unavailable.
2. Phase picking	Executed: 3637 picks; exclusions and merging checked. Full-period inference remains unfinished.
3. Training / S3–S4	Retraining not run. STEAD short links return 403; local labels and exact selections/splits missing. Original/retrained inference comparison executed on unlabeled CSOB/CMIS windows, not the exact S4 benchmark.
4. Association	Executed: 9 provisional events with an alternative external profile. Exact cited numerical starting profile missing; no author event seeds.
5. VELEST	Generic software compiled. 3 geometry-screened candidates do not resolve the chosen layered model/corrections; inversion not run.
6. TomoDD	Not run. Independent calibrated input catalog and complete differential-time/configuration workflow missing.
7. HypoDD SVD	Not run. Depends on independently relocated regional subsets and fitted model; generic software source accessible.
8. Magnitudes	Measurements executed; 0 qualifying pair equations, 0 calibrated magnitudes. Routine Md permission is resolved.
9. Completeness / b	Not calculated: no independently calibrated magnitude sample.

Stage	Actual execution and limitation
10. Focal mechanisms	Not run. Reviewed polarities, calibrated locations/model and gCAP configuration missing. Generic software accessible. Data S3 row-count discrepancy resolved.
11. Location checks	Residuals, gaps and internal S–P checks executed. Zero associated CPOZ P arrivals; relocated sector gathers, manual deep-event review and SVD validation unfinished.
12. Hybrid search	Executed on 24.298 available hours; 4 unconfirmed candidates. Three-year classification, clustering and catalog-wide FI review unfinished.
13. Historical context	Executed original leveling-table extraction (128 entries) and external routine-catalog context. Original RITE GNSS processed through 2019 with explicit overlap alignment; daily 2020–2023 continuation missing.

The unresolved starting profile, missing acceleration access and incomplete full-period processing are distinct limitations. Lack of a full catalog is not attributed solely to the prompt. Full retrieval, qualified calibration-event selection, VELEST inversion, coherency-weighted differential times, TomoDD relocation, HypoDD uncertainty, reviewed polarities, SKHASH and gCAP inversion, complete magnitude statistics and multiyear hybrid classification remain necessary. Unknown settings require documented assumptions and sensitivity checks rather than hidden substitutions. No coupled physical model is invented.

D7 Rebuild, provenance and replacement

The README gives environment creation, exact snapshot restoration and the full analysis rebuild command. For the present presentation-only revision, run `.venv/bin/pythonscripts/build_documents.py--clean` from the repository root. This compiles both PDFs from the existing saved results and local publication PDFs without running an analysis or changing numerical outputs. Its validation record is `report/document_build.json`. The checksums in `report/layout_baseline.json` freeze the numerical outputs and analytical scripts used for this revision. The original `data/manifest.sha256` and `data/build_log.json` remain evidence of the earlier scientific build; the document build has a separate hash record. The offline rebuild verifies supplied-source and downloaded-input checksums before removing generated products. It reruns the original checks and waveform processing, extended picking/association, analytic association validation, external context extraction, response-corrected amplitude/quality measurements, template scans, plotting, panel tables and the document builder, which runs two LaTeX passes for each PDF and verifies their order. This describes the full scientific rebuild; it was not rerun for this presentation-only revision. Command outcomes are saved in `data/build_log.json`. Input hashes and placement URLs are in `docs/rebuild_inputs.json`. Live routine pages can change; restoration refuses a mismatched hash and asks for the recorded snapshot or an explicit new provenance audit. Successful HTTP access is not taken as proof of reproducibility of an older mutable archive version.

The 9 October scientific run replaced the previous report source/PDF, station/coverage figures, panel records, input policy, blocker descriptions and rebuild procedure. It preserved prior observational template figures, raw source documents, independent C++ checks and useful setup; historical erroneous products remain only in Git history. That run's outputs are saved under `data/extended/`. The 12 October presentation revision replaces the report source/PDF and adds the comparison PDF, separated sources and document-build records; its scientific files and panel classifications are unchanged. Exact

replacement details are in `docs/REPLACEMENTS.md` and `docs/REPORT_LAYOUT.md`. Neither build declares the scientific project complete.

All project inputs, environments and intermediate products reside under this repository in `inputs/`, `.venv/`, `tmp/` and `build/`. Standard dependency caches may reside in `/home/mehffin/.cache/pip`; the earlier initial SeisBench configuration remains in `/home/mehffin/.seisbench`. Current analysis redirects SeisBench, plotting and numba caches into the repository. Large inputs and generic software downloads are not committed. Their URLs, sizes, checksums or pinned revisions accompany the project.

D8 Figure-by-figure record

The following table records every publication panel or labeled view. For every panel marked not reproduced, no command generates a claimed reproduction; null command fields in the machine-readable record mean unexecuted, not unavailable software. The complete per-panel input paths and command fields are in `docs/panel_record.json`. Each listed external file's exact URL, size and checksum appears in `docs/rebuild_inputs.json`. The partial panels use the recipes below, from the repository root. Each script is invoked with `.venv/bin/pythonscripts/NAME.py`; required upstream outputs are produced by `scripts/rebuild.py--clean`.

- 1B: data `context_data.py`; plot `plot_extended.py`. Inputs: `inputs/extended/delgaudio2010_table.doc`, `inputs/extended/demartino_supp.zip` (member `RITE_daily.txt`), and `inputs/extended/monthly_2023_12.pdf`.
- S1: data `analyze_observations.py` and `extended_analysis.py`; plot `plot_extended.py`. Inputs: all 171 miniSEED paths explicitly listed in the S1 record and frozen manifest. Actual recorded segments supply the plotted intervals.
- S2: data `analyze_observations.py` and `extended_analysis.py`; plot `plot_extended.py`. Inputs: `inputs/models/stations.csv`, `inputs/metadata/channels.txt`, and the 26 XML response files explicitly listed in the S2 record under `inputs/extended/responses/`.
- S15C: data `analyze_observations.py`; plot `plot_observations.py`. Inputs: `inputs/waveforms/E3_CAAM.mseed`, `inputs/waveforms/E3_COLB.mseed`, and `inputs/metadata/response.xml`.

External context recipes for 1A/C and S5 use `context_data.py` and `plot_extended.py`; the input routine HTML pages are identified row by row in the saved external CSV. Additional diagnostics use `compare_pickers.py`, `location_consistency.py`, or `review_hybrid_candidates.py`, with saved numerical outputs. These commands do not upgrade the corresponding unreproduced publication panels.

Panel/view	Quantity	Status, specification and limitation
1A	INGV event map and large-event markers	not reproduced. Main p.71, Fig.1 caption. 2005-01-01 to 2025-03-20; origin 14.14 E, 40.82 N; sea-level depth; Md>=4 markers. External routine-catalog context is plotted and labeled separately. It is not an independently detected historical catalog or a reproduced analysis result.
1B	Vertical displacement at Pozzuoli port	partially reproduced. Main p.71, Fig.1 caption. 1905-2023; Rione Terra. Original leveling table independently extracted and referenced to March 1905; 128 entries through July 2009. Original RITE GNSS data through 2019 are independently aligned using overlapping leveling epochs; 2020–2023 daily observations remain missing.

Panel/view	Quantity	Status, specification and limitation
1C	Temporal seismicity and cumulative counts	not reproduced. Main p.71, Fig.1 caption. 2005-2025; cumulative Md>0.2; study interval 2022-01-21 to 2025-03-20. External routine-catalog context is plotted and labeled separately. It is not an independently detected historical catalog or a reproduced analysis result.
2A	Catalog seismicity rates by region	not reproduced. Main p.72, Fig.2 caption. Central vs ring/other regions; 54319 vs 12083 catalog events; time bin unspecified. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted.
2B	Depth versus time, symbol size by magnitude	not reproduced. Main p.72, Fig.2 caption. Sea-level depth; near-surface 0-0.5 km; central 0.5-2.1; ring sectors to 3.7 km. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted.
3/map	Epicentral spatial pattern	not reproduced. Main p.73, Fig.3 caption. Relative origin 14.14 E, 40.82 N; sea-level depths. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted.
3/northing-depth	Depth versus northing	not reproduced. Main p.73, Fig.3 caption. Relative origin 14.14 E, 40.82 N; sea-level depths. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted.
3/ easting-depth	Depth versus easting	not reproduced. Main p.73, Fig.3 caption. Relative origin 14.14 E, 40.82 N; sea-level depths. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted.
4A	Ring-sector focal mechanisms and seismicity	not reproduced. Main p.74, Fig.4; supplement S6-S7. High quality: SKHASH A or Kagan angle<30 deg; gCAP preference with stated exception. Independent calibrated locations, reviewed polarities and gCAP configurations are missing; focal inversions not run. Data S3 actually contains 74 rows/58 gCAP entries, consistent with methods; caption says 55.
4B	Central/onshore focal mechanisms and seismicity	not reproduced. Main p.74, Fig.4; supplement S6-S7. High quality: SKHASH A or Kagan angle<30 deg; gCAP preference with stated exception. Independent calibrated locations, reviewed polarities and gCAP configurations are missing; focal inversions not run. Data S3 actually contains 74 rows/58 gCAP entries, consistent with methods; caption says 55.
S1	Station data-availability timeline	partially reproduced. Supplement S12 caption; S2 methods. 48 stations; continuous-data availability. Verified retrieved channel intervals only; full-period network collection remains unfinished and RAN credentials are unavailable.
S2	Station map	partially reproduced. Supplement S13 caption; S2 methods. 23 broadband, 9 short-period, 16 accelerometers. 51-station permitted metadata inventory differs from the paper's 48-station description; exact epoch/membership and basemap not reproduced.

Panel/view	Quantity	Status, specification and limitation
S3A	Noise example: waveform, phase labels, spectrogram	not reproduced. Supplement S14 caption; S2-S3. 30-s windows; P/S labels; D amplitude rescaling. Exact training traces and labels not identified from permissible metadata. Retraining was not repeated; permitted trained weights used for inference only.
S3B	Single-event example: waveform, phase labels, spectrogram	not reproduced. Supplement S14 caption; S2-S3. 30-s windows; P/S labels; D amplitude rescaling. Exact training traces and labels not identified from permissible metadata. Retraining was not repeated; permitted trained weights used for inference only.
S3C	Two overlapping events: waveform, phase labels, spectrogram	not reproduced. Supplement S14 caption; S2-S3. 30-s windows; P/S labels; D amplitude rescaling. Exact training traces and labels not identified from permissible metadata. Retraining was not repeated; permitted trained weights used for inference only.
S3D	Five overlapping copies of one event: waveform, phase labels, spectrogram	not reproduced. Supplement S14 caption; S2-S3. 30-s windows; P/S labels; D amplitude rescaling. Exact training traces and labels not identified from permissible metadata. Retraining was not repeated; permitted trained weights used for inference only.
S4A	Swarms at CSOB: waveform, spectrogram, original and retrained picker scores	not reproduced. Supplement S15 caption; S2-S3. 120 s; ~2 h after May 2024 Md4.4 event; phase threshold 0.3. Original and retrained models both executed on independently selected CSOB/CMIS calendar windows. Exact published windows and independent reference annotations remain unidentified; these diagnostics are not the published benchmark.
S4B	CMIS noise: waveform, spectrogram, original and retrained picker scores	not reproduced. Supplement S15 caption; S2-S3. 1600 s; coastal noise; phase threshold 0.3. Original and retrained models both executed on independently selected CSOB/CMIS calendar windows. Exact published windows and independent reference annotations remain unidentified; these diagnostics are not the published benchmark.
S5/map	Routine-catalog spatial views; map	not reproduced. Supplement S16 caption. 12083 events; 1D model and HYPO71. External routine-catalog context is plotted and labeled separately. It is not an independently detected historical catalog or a reproduced analysis result.
S5/northing-depth	Routine-catalog spatial views; northing-depth	not reproduced. Supplement S16 caption. 12083 events; 1D model and HYPO71. External routine-catalog context is plotted and labeled separately. It is not an independently detected historical catalog or a reproduced analysis result.
S5/ easting-depth	Routine-catalog spatial views; easting-depth	not reproduced. Supplement S16 caption. 12083 events; 1D model and HYPO71. External routine-catalog context is plotted and labeled separately. It is not an independently detected historical catalog or a reproduced analysis result.
S6/map	Initial-location spatial views; map	not reproduced. Supplement S17 caption. 61809 events; five VELEST runs. Generic VELEST compiled. The small independent candidate set is insufficient to resolve the chosen layered velocities and station corrections; paper uses 373 calibration events. Inversion not executed.
S6/northing-depth	Initial-location spatial views; northing-depth	not reproduced. Supplement S17 caption. 61809 events; five VELEST runs. Generic VELEST compiled. The small independent candidate set is insufficient to resolve the chosen layered velocities and station corrections; paper uses 373 calibration events. Inversion not executed.

Panel/view	Quantity	Status, specification and limitation
S6/ easting-depth	Initial-location spatial views; easting-depth	not reproduced. Supplement S17 caption. 61809 events; five VELEST runs. Generic VELEST compiled. The small independent candidate set is insufficient to resolve the chosen layered velocities and station corrections; paper uses 373 calibration events. Inversion not executed.
S7A/map	Cluster locations and 2-sigma errors: map and cross-sectional views; map	not reproduced. Supplement S18 caption; S5 methods. Central174, north165, southwest175, east166; HypoDD SVD; Table S1. No independent relocated catalog or fitted 1D model for regional HypoDD SVD tests. Generic HypoDD source is accessible; scientific runs are unfinished.
S7A/ easting-depth	Cluster locations and 2-sigma errors: map and cross-sectional views; easting-depth	not reproduced. Supplement S18 caption; S5 methods. Central174, north165, southwest175, east166; HypoDD SVD; Table S1. No independent relocated catalog or fitted 1D model for regional HypoDD SVD tests. Generic HypoDD source is accessible; scientific runs are unfinished.
S7B	EX uncertainty versus differential-pick count	not reproduced. Supplement S18 caption. Log-log axes; NCCP+NCCS. No independent relocated catalog or fitted 1D model for regional HypoDD SVD tests. Generic HypoDD source is accessible; scientific runs are unfinished.
S7D	EY uncertainty versus differential-pick count	not reproduced. Supplement S18 caption. Log-log axes; NCCP+NCCS. No independent relocated catalog or fitted 1D model for regional HypoDD SVD tests. Generic HypoDD source is accessible; scientific runs are unfinished.
S7F	EZ uncertainty versus differential-pick count	not reproduced. Supplement S18 caption. Log-log axes; NCCP+NCCS. No independent relocated catalog or fitted 1D model for regional HypoDD SVD tests. Generic HypoDD source is accessible; scientific runs are unfinished.
S7C	EX uncertainty density by cluster	not reproduced. Supplement S18 caption. Kernel density estimate; bandwidth unspecified. No independent relocated catalog or fitted 1D model for regional HypoDD SVD tests. Generic HypoDD source is accessible; scientific runs are unfinished.
S7E	EY uncertainty density by cluster	not reproduced. Supplement S18 caption. Kernel density estimate; bandwidth unspecified. No independent relocated catalog or fitted 1D model for regional HypoDD SVD tests. Generic HypoDD source is accessible; scientific runs are unfinished.
S7G	EZ uncertainty density by cluster	not reproduced. Supplement S18 caption. Kernel density estimate; bandwidth unspecified. No independent relocated catalog or fitted 1D model for regional HypoDD SVD tests. Generic HypoDD source is accessible; scientific runs are unfinished.
S8A	Estimated local versus routine duration magnitude	not reproduced. Supplement S19 caption; S5-S6 methods. Relative-amplitude LS; SNR>=5 dB; >=2 common stations; separation<distance/5. Independent amplitude/SNR measurements executed but no qualifying relative-magnitude edge; zero calibrated magnitudes. INGV Md is now a permitted required calibration reference; no catalog magnitudes fill missing results.
S8B	Magnitude-frequency distribution and fitted b value	not reproduced. Supplement S19 caption; S6 methods. Published Mc=0.4 by maximum curvature; b=0.97 by ML; bin size unspecified. Independent amplitude/SNR measurements executed but no qualifying relative-magnitude edge; zero calibrated magnitudes. INGV Md is now a permitted required calibration reference; no catalog magnitudes fill missing results.

Panel/view	Quantity	Status, specification and limitation
S9/map	Filtered-catalog spatial views and feature labels A-D; map	not reproduced. Supplement S20 caption. 38103 events; NCCP+NCCS threshold 100. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted.
S9/northing-depth	Filtered-catalog spatial views and feature labels A-D; northing-depth	not reproduced. Supplement S20 caption. 38103 events; NCCP+NCCS threshold 100. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted.
S9/ easting-depth	Filtered-catalog spatial views and feature labels A-D; easting-depth	not reproduced. Supplement S20 caption. 38103 events; NCCP+NCCS threshold 100. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted.
S10/2022	Annual seismicity map	not reproduced. Supplement S21 caption. Year 2022. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted.
S10/2023	Annual seismicity map	not reproduced. Supplement S21 caption. Year 2023. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted.
S10/2024	Annual seismicity map	not reproduced. Supplement S21 caption. Year 2024. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted.
S10/2025	Annual seismicity map	not reproduced. Supplement S21 caption. Year 2025 through March20. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted.
S11/swarm-1	Map of one of four major swarms	not reproduced. Supplement S22 caption. 2022-2025; exact swarm windows not enumerated in caption. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted.
S11/swarm-2	Map of one of four major swarms	not reproduced. Supplement S22 caption. 2022-2025; exact swarm windows not enumerated in caption. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted.
S11/swarm-3	Map of one of four major swarms	not reproduced. Supplement S22 caption. 2022-2025; exact swarm windows not enumerated in caption. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted.

Panel/view	Quantity	Status, specification and limitation
S11/swarm-4	Map of one of four major swarms	not reproduced. Supplement S22 caption. 2022-2025; exact swarm windows not enumerated in caption. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted.
S12A	Ring-sector division	not reproduced. Supplement S23 caption; S8. East, southwest, north; sector boundaries by location/depth. CPOZ observations retrieved, but this provisional catalog contains zero independently associated CPOZ P arrivals. Internal S-P consistency calculated at other stations; relocated sector gathers and location validation remain unfinished.
S12B	Ring-sector division in complementary spatial view	not reproduced. Supplement S23 caption; S8. East, southwest, north; sector boundaries by location/depth. CPOZ observations retrieved, but this provisional catalog contains zero independently associated CPOZ P arrivals. Internal S-P consistency calculated at other stations; relocated sector gathers and location validation remain unfinished.
S12C	Common-receiver gather: east / HHZ	not reproduced. Supplement S23 caption. CPOZ; -1 to +7 s about P; >4 Hz; north sector ordered by easting, others northing. CPOZ observations retrieved, but this provisional catalog contains zero independently associated CPOZ P arrivals. Internal S-P consistency calculated at other stations; relocated sector gathers and location validation remain unfinished.
S12D	Common-receiver gather: southwest / HHZ	not reproduced. Supplement S23 caption. CPOZ; -1 to +7 s about P; >4 Hz; north sector ordered by easting, others northing. CPOZ observations retrieved, but this provisional catalog contains zero independently associated CPOZ P arrivals. Internal S-P consistency calculated at other stations; relocated sector gathers and location validation remain unfinished.
S12E	Common-receiver gather: north / HHZ	not reproduced. Supplement S23 caption. CPOZ; -1 to +7 s about P; >4 Hz; north sector ordered by easting, others northing. CPOZ observations retrieved, but this provisional catalog contains zero independently associated CPOZ P arrivals. Internal S-P consistency calculated at other stations; relocated sector gathers and location validation remain unfinished.
S12F	Common-receiver gather: east / HHN	not reproduced. Supplement S23 caption. CPOZ; -1 to +7 s about P; >4 Hz; north sector ordered by easting, others northing. CPOZ observations retrieved, but this provisional catalog contains zero independently associated CPOZ P arrivals. Internal S-P consistency calculated at other stations; relocated sector gathers and location validation remain unfinished.
S12G	Common-receiver gather: southwest / HHN	not reproduced. Supplement S23 caption. CPOZ; -1 to +7 s about P; >4 Hz; north sector ordered by easting, others northing. CPOZ observations retrieved, but this provisional catalog contains zero independently associated CPOZ P arrivals. Internal S-P consistency calculated at other stations; relocated sector gathers and location validation remain unfinished.

Panel/view	Quantity	Status, specification and limitation
S12H	Common-receiver gather: north / HHN	not reproduced. Supplement S23 caption. CPOZ; -1 to +7 s about P; >4 Hz; north sector ordered by easting, others northing. CPOZ observations retrieved, but this provisional catalog contains zero independently associated CPOZ P arrivals. Internal S-P consistency calculated at other stations; relocated sector gathers and location validation remain unfinished.
S13A	Full caldera seismicity on satellite reference	not reproduced. Supplement S24 caption. Depth-coded relocated events; Google Earth basemap. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted.
S13B	Onshore seismicity on satellite reference	not reproduced. Supplement S24 caption. Depth-coded relocated events; Google Earth basemap. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted.
S14/1	CAAM waveforms and event annotations	not reproduced. Supplement S25 caption; S9-S11. 1-10 Hz display; 8-s FI at 2-5 and 8-15 Hz; FI<0 colored red. All supported downloaded CAAM/COLB intervals were scanned; exposure is about one day, not three years. Four candidate review examples remain unconfirmed. Published cluster histories and cluster-relative magnitudes are not reproduced.
S14/2	CAAM waveforms and event annotations	not reproduced. Supplement S25 caption; S9-S11. 1-10 Hz display; 8-s FI at 2-5 and 8-15 Hz; FI<0 colored red. All supported downloaded CAAM/COLB intervals were scanned; exposure is about one day, not three years. Four candidate review examples remain unconfirmed. Published cluster histories and cluster-relative magnitudes are not reproduced.
S14/3	CAAM waveforms and event annotations	not reproduced. Supplement S25 caption; S9-S11. 1-10 Hz display; 8-s FI at 2-5 and 8-15 Hz; FI<0 colored red. All supported downloaded CAAM/COLB intervals were scanned; exposure is about one day, not three years. Four candidate review examples remain unconfirmed. Published cluster histories and cluster-relative magnitudes are not reproduced.
S14/ unclustered	CAAM waveforms and event annotations	not reproduced. Supplement S25 caption; S9-S11. 1-10 Hz display; 8-s FI at 2-5 and 8-15 Hz; FI<0 colored red. All supported downloaded CAAM/COLB intervals were scanned; exposure is about one day, not three years. Four candidate review examples remain unconfirmed. Published cluster histories and cluster-relative magnitudes are not reproduced.
S15A	Hybrid family 1: waveforms and spectrograms	not reproduced. Supplement S26 caption; S9-S10. CAAM/COLB HHZ/HHE; amplitudes normalized by CAAM maximum. Post-generation visual comparison shows a different published example from the written-method template used here. The generated family-template figure is related analysis, not reproduction of this panel.
S15B	Hybrid family 2: waveforms and spectrograms	not reproduced. Supplement S26 caption; S9-S10. CAAM/COLB HHZ/HHE; amplitudes normalized by CAAM maximum. Post-generation visual comparison shows a different published example from the written-method template used here. The generated family-template figure is related analysis, not reproduction of this panel.

Panel/view	Quantity	Status, specification and limitation
S15C	Hybrid family 3: waveforms and spectrograms	partially reproduced. Supplement S26 caption; S9-S10. CAAM/COLB HHZ/HHE; amplitudes normalized by CAAM maximum. Written template times used; exact published example alignment, display filtering and spectral settings incompletely specified. Sensitivity-corrected velocities and stated independent display settings used.
S15D	Shallow volcano-tectonic example: waveforms and spectrograms	not reproduced. Supplement S26 caption; S9-S10. CAAM/COLB HHZ/HHE; amplitudes normalized by CAAM maximum. Exact shallow VT example not identified by written method alone; no author catalog or plot values used to select it.

D9 Saved numerical tables and equation consistency checks

Original-waveform template frequency indices

These saved template measurements are supporting evidence, not a reproduction of the complete S14 cluster analysis. E1 and E2 are written-method templates, not the exact S15A/B illustrated events.

Template	FI	Samples
E1	-0.7862	800
E2	-0.4286	800
E3	-1.0819	800

Analytic fixtures

Check	Absolute error	Tolerance
equal_amplitudes	0	1e-14
decade_amplitude	0	1e-14
ratio_reversal	0	1e-14
common_station_gain	5.55e-17	1e-14
zero_amplitude_rejected	0	0
negative_power_rejected	0	0
fi_mixture_0	4e-15	1e-10
fi_mixture_1	8.88e-16	1e-10
fi_mixture_2	1.78e-15	1e-10
fi_mixture_3	1.55e-15	1e-10
fi_mixture_4	1.78e-15	1e-10
fi_mixture_5	2.11e-15	1e-10
fi_mixture_6	2.22e-16	1e-10
fi_mixture_7	2.11e-15	1e-10
fi_mixture_8	1.33e-15	1e-10
fi_mixture_9	8.88e-16	1e-10
fi_mixture_10	1.33e-15	1e-10

Check	Absolute error	Tolerance
fi_mixture_11	4.44e-16	1e-10
fi_mixture_12	1.78e-15	1e-10
parseval	3.33e-15	1e-10
fi_circular_shift	7.22e-16	1e-10
fi_gain	8.88e-16	1e-10
snr_5s_2s_windows	8.88e-16	1e-10
magnitude_exact_recovery	1.94e-16	1e-12
magnitude_pair_order	0	1e-12
magnitude_gauge_shift	4.44e-16	1e-12
unanchored_problem_rejected	0	0

D10 Sources

Primary specification: Tan et al. (2025), Science 390, 70–75, local `tanmodel.pdf` and `tanmodelsupp.pdf`. Original waveform and response source: https://webservices.ingv.it/fdsnws/. Required routine magnitude reference and external context: https://terremoti.ov.ingv.it/gossip/flegrei/. Original leveling table: Del Gaudio et al. (2010), doi:10.1016/j.jvolgeores.2010.05.014, publisher supplement `1-s2.0-S0377027310001745-mmc1.doc`. External background references: Battaglia et al. (2008), doi:10.1111/j.1365-2478.2007.00687.x; Calò and Tramelli (2018), doi:10.1038/s41598-018-34456-x. Trained artifact and candidate fixed background release: https://github.com/iamharrison2000/Campi-Flegrei-Earthquake-Monitoring, revision pinned in the receipts. GaMMA: https://github.com/AI4EPS/GaMMA, revision `80394dd4a4c29450da599f507916ecf1db6203dd`. VELEST: https://seg.ethz.ch/software/velest.html. Additional source receipts record RAN, temporary observations, STEAD, HypoDD, SKHASH and gCAP. No Tan study-specific analysis or plotting script was used.